

Chapter 23

New Dimensions of Hillslope Hydrology

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23.1 Introduction

Hillslopes are fundamental landscape units that strongly control the processes whereby precipitation or snowmelt is vertically and laterally transported to the streams. They are, in a sense, microcosms of catchments. Understanding and predicting the hillslope response is highly important in terms of flood prediction, transport of nutrients and sediments into surface water bodies, slope stability, and soil-atmosphere-vegetation exchange processes. Through numerous field experiments and numerical studies, much progress has been made in hillslope hydrology in the past decades. However, our ability to extrapolate these findings to ungauged hillslopes and catchments is still very poor (Sivapalan 2005). A common thread that has evolved recently is to search for the underlying principles of hydrological processes instead of characterizing and cataloging the enormous heterogeneity and complexity of rainfall-runoff processes (McDonnell et al. 2007). The aim of this chapter is to provide an overview on today's conceptual models of processes at the hillslope scale and to examine the factors driving these mechanisms. The focus will be on compiling recent findings on the dominant controls of hillslope runoff processes to meet the need of identifying underlying principles. The chapter closes with thoughts on new dimensions and directions of hillslope hydrology and research avenues to follow for the future.

23.2 Conceptual Process Models at the Hillslope Scale

Much of the process understanding in hillslope hydrology in forested landscapes has evolved from only a few intensively studied hillslopes in the Panola Mountain Research Watershed, USA (e.g., Burns et al. 1998; Freer et al. 2002; Tromp-van Meerveld and McDonnell 2006a–c), in Maimai, New Zealand (e.g., Mosley 1979; Sklash et al. 1986; McDonnell 1990; Woods and Rowe 1996), and in the Fudoji catchment, Japan (Asano et al. 2002, 2004; Uchida et al. 2003a, b, 2004). Interestingly, some of the earlier conceptual models derived from these hillslopes were later discarded through novel measurement techniques (McGlynn et al. 2002;

Tromp-van Meerveld et al. 2007). Until today, a number of experimental hillslope studies have been conducted in different environments, leading to an advanced and refined perception of rainfall-runoff processes (e.g., Scherrer et al. 2007; Graham 2008; Kienzler and Naef 2008a, b). Despite the large variety of observed flow processes, one commonality seems to be the strong nonlinearity of hillslope response to rainfall (McDonnell 2003). By definition, a system is nonlinear if the outputs are not proportional to the inputs across the entire range of the inputs (Phillips 2003). Several hillslope studies report on a certain precipitation threshold until significant hillslope outflow is triggered (e.g., Uchida et al. 2005; Tromp-van Meerveld and McDonnell 2006a; Graham 2008). Such a threshold response can be regarded as an extreme form of nonlinearity (Zehe and Sivapalan 2009). To predict this nonlinear system behavior, we need a sound concept of hydrological processes taking place at the hillslope scale.

In general, the hillslope runoff processes can be conceptualized as (1) overland flow (Hortonian and saturation overland flow [SOF]), (2) near-surface flow, (3) subsurface stormflow (SSF), and (4) deep percolation (DP) (see Fig. 23.1) (Scherrer and Naef 2003; Schmocker-Fackel 2004; Sidle et al. 2007). Once precipitation enters the hydrologic system as rainfall or snow, it will be partitioned into different flowpaths depending on the storm and hillslope characteristics.

Hortonian overland flow (HOF), also called infiltration excess overland flow, is lateral flow along the surface when the rainfall intensity exceeds the infiltration capacity of the soil (Horton 1933). Factors fostering HOF are surface sealing and crust building, soil compaction, an abrupt increase of bulk density, topsoil hydrophobicity, and low macroporosity (e.g., Doerr et al. 2000; Scherrer and Naef 2003; Weiler and Naef 2003; Schmocker-Fackel 2004). SOF takes place once the storage capacity of the soil is exceeded and no further infiltration is possible.

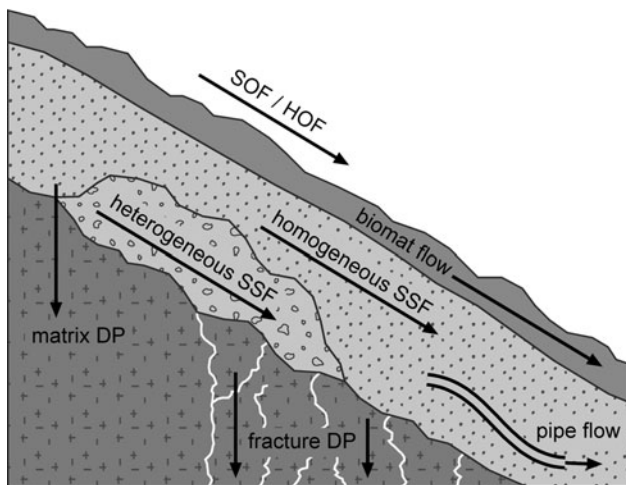


Fig. 23.1 Conceptual process models at the hillslope scale

SOF is thus dependent on antecedent wetness and preferential flow paths, transporting water into deeper soil layers. Once surface flow is established, runoff pathways are influenced by surface roughness (e.g., microrelief, stone and vegetation cover) (Bracken and Croke 2007), reinfiltration and run-on processes (Corradini et al. 1998). True HOF has been rarely observed on forested hillslopes (Sidle et al. 2007), except for HOF generated by disturbed forest soils due to skid roads, trails, and log-landing decks. When overland flow occurs on forest slopes, it is frequently believed to result from return flow (Dunne and Black 1970), which occurs when subsurface water is forced back to the surface (e.g., due to a change in slope or low permeability layers) (Schmocker-Fackel 2004; Sidle et al. 2007), or from saturated areas (Elsenbeer and Vertessy 2000; Godsey et al. 2004). Experimental studies measuring surface runoff via a collection trough, which is usually inserted below the soil surface, often mistakably assume to catch HOF, even though it in fact originates from flow within the forest litter layer (Buttle and Turcotte 1999; Sidle et al. 2007). This near-surface flow phenomenon has been called “organic layer interflow” (Weiler and McDonnell 2004), “biomat flow” (Sidle et al. 2007), “pseudo-overland flow” (McDonnell et al. 1991), and “thatched roof” effect (Ward and Robinson 1990).

The conceptualization of flow processes in the subsurface is more complex. Generally, quickly responding SSF and DP can be discriminated. SSF, also termed subsurface flow, interflow, throughflow, lateral flow, and shallow groundwater flow, is the fast lateral movement of water through the soil matrix or permeable bedrock above a layer of reduced permeability (Dunne 1978; Weiler et al. 2006). DP or groundwater recharge occurs when infiltrated water vertically seeps into the bedrock, thus contributing to slow baseflow (Scherrer et al. 2007). In many early studies on hillslope hydrology, the underlying bedrock was assumed impermeable. However, recently published work showed that both infiltration into the bedrock and exfiltration from the bedrock into the soil can be significant and hence cannot be ignored (e.g., Anderson et al. 1997; Uchida et al. 2002; Kosugi et al. 2006; Tromp-van Meerveld et al. 2007; Graham 2008; Tromp-van Meerveld and Weiler 2008).

The complex nature of SSF formation is subject of ongoing research. A prerequisite for SSF is a transient saturation of parts of the soil matrix (≥ 0 kPa matrix potential) due to a layer of reduced permeability in vertical direction (Weiler et al. 2006). This can be the bedrock or low permeability soil horizons due to a change in soil texture, an increase in bulk density, or a decrease in macroporosity (Weiler et al. 2006). Such low permeability soil horizons develop from different pedogenetic processes (e.g., *lixiviation* leading to *argic* horizons), biological processes (reduced earthworm activity or decreasing rooting density), or due to management practices (plough layer, landfill lining). After the soil matrix has reached partial saturation, SSF can take place through high permeability layers of the soil matrix or through soil pipes. High permeability layers are coarse textured and/or have a large pore space. They are often found in talus slopes, landslide debris, periglacial solifluction deposits, or unconsolidated moraine material (Weiler et al. 2006). Soil pipes are defined as lateral voids such as decayed roots, animal burrows, channels due to subsurface hydraulic erosion, and tension cracks (Tsukamoto et al. 1988). They are mostly observed at the soil–bedrock interface (or, more

generally, impeding layer interface), but also within the soil profile and at the organic horizon-mineral soil interface (Sidle et al. 2001; Holden and Burt 2002; Weiler et al. 2006). Soil pipes are mainly short and discontinuous features; however, they were found to form a network of interconnected pipes, which develops parallel to the slope gradient (Terajima et al. 2000; Sidle et al. 2001; Uchida et al. 2001). The interconnection of pipes is facilitated by saturated parts of the soil matrix, high-permeability areas such as pockets of organic material and loose soil, and water emanating from bedrock fractures (Noguchi et al. 1999; Sidle et al. 2001; Anderson et al. 2008). The exact mechanisms of pipe flow activation, pipe flow inter-connection, and exchange processes between pipes and the surrounding soil matrix are still not well known. Nevertheless, the significance of pipe flow making up a large fraction of total SSF was shown by numerous experimental studies (e.g., Jones 1997; Holden and Burt 2002; Uchida et al. 2002, 2005; Kienzler and Naef 2008a; Anderson et al. 2009). Pipe networks effectively drain the hillslope, increasing peak discharge and decreasing peak lag time of the storm hydrograph (Uchida et al. 2001). On the one hand, pipe flow may increase slope stability, counteracting a positive pore water pressure development of the soil matrix. On the other hand, when the capacity of pipe flow exceeded or pipes become clogged, slope failure may be triggered (Uchida et al. 2001).

We propose that SSF predominantly occurs via lateral preferential flow paths as opposed to sheet-like matrix flow, due to the high spatial variability of controlling factors (e.g., bedrock topography, variable soil depth). We define a lateral preferential flow path as either flow through high-permeability areas of the soil matrix or flow through interconnected pipes. This is similar to the concept of vertical flow processes at the plot scale, where a distinction is made between (a) homogeneous matrix flow (homogenous wetting front parallel to the soil surface), (b) heterogeneous matrix flow (preferential flow resulting from instabilities in the wetting front or from funneling due to less permeable zones), and (c) macropore flow (Hendrickx and Flury 2001; Weiler and Flühler 2004). An indication for subsurface flow via lateral preferential flow paths is a hydraulically limited water table response (e.g., Fannin et al. 2000; Anderson 2008). Once the water table rises into a zone where a lateral preferential flow network is present, the hydraulic conductivity significantly increases. After the activation of the preferential flow network, the water table remains relatively constant while the discharge continues to increase (Anderson 2008).

Depending on the pathway incoming precipitation takes (HOF, SOF, biomat flow, SSF, DP) hillslope response parameters such as response time, water transit time and the event to pre-event water ratio will strongly differ (event or “new” water = water from the current rain or snowmelt event; pre-event or “old water” = water that existed in the catchment prior to the current rain or snowmelt event (Sklash 1990); transit time = time between when a water molecule enters the hydrologic system until it exits it). While DP is characterized through mainly pre-event water and long transit times, HOF and SOF have shorter transit times and a high fraction of event water. Due to the heterogeneity of SSF processes (degree of matrix flow vs. pipe flow, timing of pipe network activation, etc.), a diversity of response characteristics for SSF is possible. Recently, Kienzler and Naef (2008a)

showed through an intercomparison study that variations in response time and event to pre-event water ratio can be explained through different degrees of direct or indirect feeding of preferential flow paths: When precipitation feeds directly into preferential flow paths, bypass flow occurs, resulting in a quick SSF response and high event water fractions. In contrast, when preferential pathways are fed indirectly via saturated parts of the soil, SSF response is delayed and contains less event water.

Albeit this progress in generalizable process understanding, predicting the hydrological response of ungauged basins remains anything but trivial. Nonetheless, there have been a few approaches trying to provide a systematic basis for inferring rainfall runoff responses, such as the UK's hydrology of soil type (HOST) classification system (Boorman et al. 1995), the decision tree approach by Scherrer and Naef (2003) to indicate hydrological processes on temperate grassland, or the field instruction for assessing surface runoff coefficients for alpine soil-/vegetation units by Markart et al. (2004). As starting point for new attempts classifying the hillslope response to rainfall, it seems promising to summarize recent findings on the dominant controls of runoff processes.

23.3 Controlling Factors of Flow Processes

Generally, the controlling factors of hillslope response to rainfall or snowmelt can be broken up into static factors, at least in the sense of shorter time scales such as years (parent material, surface and bedrock topography, exposure), also termed the “hillslope configuration” (Hopp and McDonnell 2009), and dynamic factors (precipitation characteristics, soil moisture, vegetation). Despite this division, we discuss the different controlling factors from a top-down perspective, meaning from precipitation entering the system at the upper boundary to the bedrock as lower boundary, thereby highlighting the numerous interrelations among different factors.

23.3.1 *Input Characteristics*

Input characteristics (amount, intensity, and frequency of storms, characteristics of snow pack and timing of snowmelt) clearly constitute a first-order control on runoff response. Furthermore, the temporal distribution of input and patterns of evapotranspiration strongly influence the soil moisture content, which represents a key factor driving hydrological processes (Sivapalan 1993; Jost et al. 2005; McNamara et al. 2005). Concerning input amount, there seems to be a strong nonlinearity between rainfall or snowmelt input and hillslope response. Several hillslope studies documented that a certain precipitation threshold had to be exceeded until significant hillslope outflow was triggered, e.g., observed precipitation threshold of 55 mm and approx. 23 mm at the trenched Panola and the Maimai hillslope, respectively (Tromp-van Meerveld and McDonnell 2006a; Graham 2008).

The effect of rainfall intensity on runoff generation seems rather complex. Obviously, overland flow is positively correlated with rainfall intensity (Horton 1933). Furthermore, an increase in rainfall intensity enhances macropore flow since the soil water pressure is closer to saturation during rainfall and larger macropores conduct water (Jarvis et al. 2007). Enhanced macropore flow, which channels water to deeper soil layers, leads to the assumption of a positive relation between rainfall intensity and subsurface flow response. This is supported by the finding of SSF velocities being highly correlated with the 1-h rainfall intensity by Anderson et al. (2009). However, Kienzler and Naef (2008b) conclude from sprinkling experiments at three grassland slopes with varying intensity and antecedent wetness that a higher intensity did not increase subsurface flow rates. They explain this with differing flow processes being triggered by rainfall intensity (during low intensity, saturation above the bedrock and thus subsoil SSF was most relevant; during higher intensity, topsoil saturation and overland flow was the dominant runoff process, Kienzler and Naef 2008b). It seems like the effect of rainfall intensity is strongly bound to other interacting factors (e.g., soil hydraulic properties, macroporosity, pipe network, antecedent wetness) and can be both highly relevant and subordinate when buffered by other controls.

23.3.2 *Vegetation*

The impact of vegetation cover on runoff processes and streamflow has long been a topic of interest, especially with regard to timber harvesting. Several paired catchment studies found that forested catchments are characterized by higher evapotranspiration and lower runoff, delayed peak flow and a longer hydrograph recession time (Schume et al. 2004; Jost et al. 2005). To open the black box of catchment scale processes, vegetation effects at smaller scales need to be examined. Generally, the precipitation signal is heavily modified by the vegetation growing on a slope. Besides an above-ground partitioning of precipitation through overstory and understory vegetation, plants also partition incoming water on the ground and belowground through the litter layer and roots. This “double-funneling” effect (aboveground and belowground) (Johnson and Lehmann 2006) strongly alters hydrologic and nutrient fluxes and leads to heterogeneous soil water dynamics (Johnson and Lehmann 2006; Liang et al. 2007). Preceding and subsequent chapters (Chaps. 18–22 and 24) provide detailed insights into the aspects discussed hereafter.

23.3.2.1 *Aboveground Partitioning*

Vegetation partitions precipitation into (1) interception (precipitation remaining on the vegetation, which is evaporated after or during rainfall), (2) stemflow (water flowing to the ground via trunks or stems), and (3) throughfall (precipitation falling through the vegetation reaching the ground) (Crockford and Richardson 2000) (see Chaps. 20 and 21). Concentrated water flow along stems and temporary storage on

forest canopies results in evaporative losses and intensity smoothing (Keim et al. 2006), and causes spatially variable patterns of soil moisture and solute deposition (e.g., Johnson and Lehmann 2006; Zimmermann et al. 2008; Levia et al. 2010). This may be highly relevant for runoff generation.

A number of studies documented the importance of stem flow on soil moisture patterns. To highlight some findings, Herwitz (1986) observed that high stem flow fluxes produced localized HOF in a tropical rainforest. Chang and Matzner (2000) found that 13.5% of the annual infiltration in a beech forest occurred within 1 m² of the tree base, an area solely representing 3% of the total forest area. More recently, Liang et al. (2007) observed that the pore water pressure at the soil–bedrock interface increased faster and to a greater extent downslope from a tree stem growing on a steep hillslope than upslope (Liang et al. 2007).

Snow accumulation and ablation rates are also strongly influenced by canopies (e.g., López-Moreno and Stähli 2008; Molotch et al. 2009). For instance, Molotch et al. (2009) observed a 29% greater snow accumulation in open-area vs. under-canopy locations. Additionally, snow ablation rates under canopies were reduced by 39%, which extended the duration of snowmelt in some cases (Molotch et al. 2009). The role of interception on soil moisture and runoff was examined by Keim and Skaugset (2003) and Keim et al. (2006). They conducted virtual experiments (numerical experiments driven by collective field intelligence, Weiler and McDonnell 2004) based on data from the Panola hillslope and field-measured throughfall rates. Their virtual experiments revealed that interception delayed the onset of SSF, lowered and delayed stormflow peaks, and decreased total flow and the runoff ratio (Keim et al. 2006). When they used simplified modeled throughfall rates (where they separated the effect of interception into evaporation and intensity smoothing), it became apparent that canopy evaporation was responsible for most of these effects, while intensity smoothing showed measurable differences only in peak SSF rate (Keim et al. 2006). This finding is highly relevant for the calibration of watershed models. “Not only would ignoring interception miss a major effect of vegetation on SSF generation, our work also shows that simply applying some fractional reduction as a scaled input signal (as is customary in watershed modeling studies) may mask important effects on peak flow response in some situations” (Keim et al. 2006). Another set of virtual experiments investigated the effect of rainfall intensity smoothing on soil water pore pressure and slope stability during high-intensity rainfall (Keim and Skaugset 2003). The outcome was that estimates of slope stability were generally greater under forest canopy than for the same hillslope without forest canopy (Keim and Skaugset 2003).

23.3.2.2 Partitioning on the Ground and Belowground

The fraction of rainfall that reaches the surface is further dispersed though funneling on the ground and belowground due to ground vegetation, litter layer coverage, and roots (Buttle and Turcotte 1999; Johnson and Lehmann 2006; Sidle et al. 2007) (see also Chaps. 22 and 24). A widespread feature of wooded ecosystems is a litter

layer covering the soil, consisting of dead leaves, twigs and other fragmented organic material (Sato et al. 2004). The litter layer strongly determines the infiltration capacity of the soil. It may act as barrier to flow, triggering surface runoff, or may favor vertical infiltration and hence SSF or groundwater recharge. Recently, the effects of leaf morphology and rainfall conditions on the moisture dynamics of litter layers were examined for two contrasting litter types (needle-leaf vs. broad-leaf) (Sato et al. 2004). The results showed that the broad-leaf litter intercepted more rainwater than the needle-leaf litter. Additionally, the water moved laterally in the broad-leaf litter layer, whereas it moved vertically down in the needle-leaf litter layer (Sato et al. 2004). However, it has also been observed that dry needle-leaf litter from spruce may exhibit hydrophobic properties and thus hinders vertical infiltration (Schume et al. 2004). Similar effects due to hydrophobicity were also discovered for dry, fur-like root layers in grassland topsoil (Scherrer et al. 2007), alpine mats (Markart et al. 2004), and burned areas (e.g., Doerr et al. 2000). Moreover, seasonal differences in vegetation cover influence the surface roughness and hence runoff production (Bracken and Croke 2007).

Belowground, trees further modify soil moisture and nutrient dynamics through preferential flow along roots (e.g., Bundt et al. 2001; Johnson and Lehmann 2006). These root-derived preferential flow pathways result from localized compaction of soil by roots and the addition of root exudates to adjacent soil (Johnson and Lehmann 2006). The important role of roots during infiltration was early suggested (Beven and Germann 1982). Numerous studies have confirmed the relevance of root induced preferential flow since then (e.g., Bundt et al. 2001; Buttle and McDonald 2002; Bachmair et al. 2009; Lange et al. 2009). As an example, dye tracer experiments in little developed volcanic ash soils documented a straight infiltration front in unvegetated soil; in forested soil, however, water was preferentially channeled (Blume et al. 2008). The effect of different species on water flow in soil has only been recently brought into focus. It is well known that each tree species develops a typical root system. Even though root architecture may differ within species as response to local conditions (soil properties or competition with others), following main root systems can be distinguished: taproot system (one vertically growing taproot dominates); heart-shaped system; sinker root system (shallow, plate-like network of lateral roots with downward sinker roots) (Polonski and Kuhn 1998; Schume et al. 2004). Nordmann et al. (2009) presented remarkable findings on the impact of different root systems on subsurface runoff. They conducted high-intensity rainfall experiments on trenched hillslope segments with similar soil and bedrock but different tree species to assess the water retention capacity of different species. A species-related trend in water retention could be observed: The most favorable retention rates were found in a mixed beech and sycamore maple stand (deep-rooting trees). Stands dominated by spruce showed the lowest retention rates (sinker roots). Nonetheless, the pedologic and geologic properties overcompensated the species effect at one site. Remarkably, subsurface flow paths under spruce mostly occurred on top of the impeding layer, whereas flow paths under beech and sycamore maple also existed within the impeding layer (Nordmann et al. 2009).

Evapotranspiration rates of forests are also linked with root systems (Schume et al. 2004; Schwärzel et al. 2009). Schume et al. (2004), for instance, investigated soil water depletion and recharge patterns in a mixed beech-spruce and a spruce stand by high-resolution time domain reflectometry (TDR) measurements. Maximum differences were observed in periods with high evaporative demand. They state that the overproportionate evapotranspiration of the mixed stand was exclusively attributable to beech, which deepened and intensified its fine-root system in mixture, while spruce rooted shallower. The mixed stand extracted a higher percentage of water from deeper soil layers than the pure stand (Schume et al. 2004).

23.3.3 Topography

As part of the “hillslope configuration,” topography undoubtedly represents a major hydrological driver, influencing soil properties, vegetation patterns, and microclimate (Anderson and Burt 1978; Burns et al. 1998; Wagener et al. 2007). Many of the effects topography exerts on hillslope hydrological processes are rather intuitive; nevertheless, a number of studies quantitatively assessed the role of different terrain properties (slope gradient, hillslope geometry, aspect, and microtopography) on runoff generation (e.g., Freer et al. 1997; Burns et al. 1998; Aryal et al. 2002; Troch et al. 2003; Berne et al. 2005; Bogaart and Troch 2006; Lyon and Troch 2007; Fujimoto et al. 2008). Most hillslopes have complex three-dimensional shapes, characterized by their profile curvature (topographic curvature along the flow line: convex, planar, concave) and contour or planform curvature (curvature of contour lines: divergent, parallel, convergent). These attributes significantly control the flow behavior and saturation along hillslopes (Aryal et al. 2002, 2005; Troch et al. 2003). For instance, Troch et al. (2003) conclude from a study, in which they presented a new mathematical formulation of homogeneous subsurface flow along complex hillslopes (hillslope-storage Boussinesq equation), that convergent hillslopes drain much slower than divergent hillslopes, due to the reduced flow domain near the outlet. Furthermore, the accumulation of moisture storage near the outlet of convergent hillslopes results in bell-shaped hydrographs, whereas fast draining divergent hillslopes produce peaked hydrographs (Troch et al. 2003). Field evidence on the effect of hillslope geometry was provided by Fujimoto et al. (2008), who compared the runoff behavior of two contrasting slopes (concave valley-head vs. planar side slope). Besides large-scale hillslope geometry, surface and bedrock microtopography affect the hillslope runoff response (Freer et al. 2002; Tromp-van Meerveld and McDonnell 2006b; Graham 2008). Tromp-van Meerveld and McDonnell (2006b) proposed a “fill-and-spill” mechanism, meaning that during a rainfall event bedrock depressions have to fill up before water spills over the bedrock ridges and contributes to hillslope outflow. An increase in slope angle thus likely decreases the volume of bedrock pools and the height of spilling barriers, thereby changing the balance between fill and spill and the timing and spatial distribution of water flow (Hopp and McDonnell 2009). Clearly, the fill

and spill process solely plays a role if there is an interaction between the bedrock and the overlying soil (e.g., transmissive soil).

Evidence on the coupling of topography and hydrologic response was also reported by Zehe and Flüßler (2001), who conducted plot-scale dye tracer experiments at different valley locations. They found that plots located at the bottom of slopes had a greater susceptibility to preferential flow than those in the upper part. An explanation is the presence of deeper, biologically more active, often finer textured soils at the foot slope (Zehe and Flüßler 2001). Holden (2009) also investigated the spatial distribution of macropore flow as a proportion of saturated hydraulic conductivity on humid-temperate slopes. He discovered that near-surface macropore flow varied according to slope position, yet spatial patterns differed among hillslopes due to differences in soil type.

An interesting approach linking hydrologic response to geomorphic properties was presented by Broxton et al. (2009). They hypothesized that water transit time is related to aspect (slope direction and exposure), since this determines the amount of energy a particular hillslope receives, which, in turn, affects a variety of factors such as the timing and intensity of snow-melt, evapotranspiration, and vegetation patterns (Broxton et al. 2009). Their experiments conducted in small, snowmelt-dominated mountainous catchments suggested that isotopic variability and estimated transit times were both related to aspect. Despite the observed correlations between hydrologic response and aspect, causality cannot be inferred. Nevertheless, their approach provides insight into the strengths of using aspect as a simple and transferable proxy for energy inputs to capture a variety of factors affecting water fluxes (Broxton et al. 2009).

23.3.4 Soil Properties

Obviously, soil properties strongly control the flowpaths incoming precipitation takes. Depending on the infiltration capacity of the soil, water is split up into surface flow or vertical percolation. The subsurface response is controlled by soil type, soil thickness, drainable porosity, soil pipes, macropores, and moisture content.

23.3.4.1 Soil Type, Soil Thickness, and Drainable Porosity

Generally, the type of soil strongly determines the hydrologic response. Certain pedogenetic processes result in certain properties, which, in turn, predefine hydrologic response (e.g., Luvisols: clay leaching leading to argic horizons, which may act as barrier to flow; Regosols: immature soils, low thickness above bedrock, thus prone to quick saturation) (Boorman et al. 1995; Zech and Hintermaier-Erhard 2002). Thus, information about the spatial distribution of soil types, which is available in many countries, provides a valuable tool for predicting hydrological

processes in ungauged regions (e.g., Tetzlaff et al. 2009). Soil thickness and drainable porosity, which is the pore space between field capacity and saturation, represent first-order controls on SSF (Weiler and McDonnell 2004; Uchida et al. 2006; Hopp and McDonnell 2009). Virtual experiments based on the Panola hillslope revealed that an increase in soil thickness led to a general reduction of the hillslope hydrologic response: SSF started later, the time to peak discharge increased, and peak discharge decreased (Hopp and McDonnell 2009). Another set of virtual experiments investigating the effect of variable soil depth showed that variations in soil thickness not only have a large influence on the spatial variation of subsurface flow but also largely control the total SSF volume (Weiler and McDonnell 2004). Clearly, the vertical travel distance of rainfall from the soil surface to the impeding layer is longer in deeper soils. Thus, the build-up of saturation and the initiation of subsurface flow occur later. Due to the larger total storage volume of deeper soils, incoming precipitation peaks are damped (Hopp and McDonnell 2009). A similar effect emanates from drainable porosity. While a rapid decline in drainable porosity with depth accelerates the development of a transient saturation, a high drainable porosity in combination with high moisture deficits causes a long lag time till SSF is triggered and large event water ratios (Uchida et al. 2004, 2006; Weiler and McDonnell 2004).

23.3.4.2 Soil Pipes and Macropores

Pipe networks seem to be an eminent feature of most hillslopes, as numerous studies suggest (e.g., Jones 1997; Holden and Burt 2002; Uchida et al. 2005; Tromp-van Meerveld and McDonnell 2006a; Graham 2008; Kienzler and Naef 2008a; Anderson et al. 2009). In consequence, the existence and nature of soil pipes, as well as factors influencing the maintenance and interconnection of pipes, govern the subsurface hillslope response to a large extent. Despite this importance, we lack a solid understanding on the controls of pipe flow. The evolution of pipe networks seems to be a complex interaction between mechanical processes (burrowing animals, roots, subsurface erosion), soil, topography, climate, and contributing area (Tsuboyama et al. 1994). Anderson et al. (2009), for instance, observed SSF velocities in a hillslope in coastal British Columbia, Canada, that were among the highest velocities reported in literature. They attribute this to the humid climate (annual precipitation >2,000 mm), shallow soils (often less than 1 m deep), and the lush vegetation, which caused a highly developed preferential flow network. Dye staining experiments and excavations have shown that pipe flow often occurs in decayed root channels (Noguchi et al. 1999; Anderson et al. 2008). Root architecture may thus play a significant role for pipe evolution. Once pipes are formed, subsurface erosion is considered to be highly important for the maintenance and modification of pipes (Uchida et al. 2001; Uchida 2004). Anderson et al. (2008) observed a link between contributing area and distribution of preferential flow features. They showed that the presence of highly developed preferential flow

features corresponded to large surface contributing areas, whereas few preferential flow paths coincided with small contributing areas and relatively flat local topography. This suggests that soils with small contributing area may not receive flow rates large enough to modify and maintain large preferential flow features (Anderson et al. 2008). Moreover, it seems like steeper slopes have better developed pipes, since higher gradients result in higher energy for pipe erosion. The interconnection of pipes, which strongly increases the effectiveness of hillslope drainage, is assumed to increase with soil moisture (Tsuboyama et al. 1994; Sidle et al. 2000; Uchida et al. 2005; Anderson et al. 2009).

Besides pipe flow, vertical macropores play an important role for hillslope runoff processes, since macropores increase the infiltration capacity and quickly transport precipitation to deeper soil layers. The knowledge on macropore flow is much further advanced than that on pipe flow (for a recent review on controlling factors of macropore flow, see Jarvis et al. (2007)).

23.3.4.3 Soil Moisture

It is self-evident that the hillslope wetness constitutes a first-order control on the timing and the amount of runoff (e.g., Sivapalan 1993; Grayson et al. 1997; Western et al. 2003). However, it is not the soil moisture state alone, it is also the connectedness of saturated soil patches that determines the runoff response (e.g., Stieglitz et al. 2003; Western et al. 2003; McNamara et al. 2005; Tromp-van Meerveld and McDonnell 2006b; James and Roulet 2007). Generally, the wetter the system, the faster and more intense the hydrologic response will be. This applies to both vertical processes such as macropore flow (Weiler 2001), and to lateral processes at the hillslope scale (Kienzler and Naef 2008b). However, Scherrer et al. (2007) inferred from hillslope sprinkling experiments at several different sites that the impact of antecedent soil moisture on runoff volume depends on the runoff process. Kienzler and Naef (2008b) explain the variable influence of antecedent wetness with different mechanisms of SSF initiation (high influence if SSF is initiated via saturated soil patches vs. low influence when SSF is directly fed by preferential pathways).

Soil moisture conditions are characterized by strong seasonality, which not only results from meteorological conditions but also from vegetation dynamics and thus transpiration (e.g., Jackson et al. 2000; Western et al. 2003; Schume et al. 2004; McNamara et al. 2005; Williams et al. 2008; Kim 2009). McNamara et al. (2005) identified five soil moisture states throughout the year governing the runoff response in a snowmelt-driven, semiarid catchment: (1) a summer dry period (soil water depletion due to evaporation exceeding precipitation), (2) a transitional fall wetting period, (3) a winter wet, low-flux period (precipitation falling as snow), (4) a spring wet, high-flux period (snowmelt), and (5) a transitional late-spring drying period. Despite seasonal variations, soil moisture patterns are spatially highly variable, which can be attributed to topography and vegetation effects, as highlighted before.

23.3.5 *Geologic Properties*

Bedrock topography was already highlighted as a control on runoff generation (fill-and-spill hypothesis by Tromp-van Meerveld and McDonnell 2006b). Evidence supporting this hypothesis was also presented by Graham (2008). During hillslope excavations in the Maimai catchment, he observed that flow paths along the soil-bedrock interface were routed primarily by features such as protruding cobbles and rills on the bedrock surface (Graham 2008). Apart from bedrock topography, the bedrock permeability mainly controls how much water can vertically percolate and laterally redirected, respectively. Tromp-van Meerveld and Weiler (2008) criticize how many experimental hillslope studies and hillslope models assume impermeable bedrock and thus no-flow boundary conditions at the soil–bedrock interface, despite clear evidence of vertical leaching into the bedrock (e.g., Kosugi et al. 2006; Tromp-van Meerveld et al. 2007; Graham 2008; Kosugi et al. 2008b). For instance, a falling head permeability test recently indicated that the bedrock underlying the Maimai hillslope is semipervious (point observation of saturated hydraulic conductivity 2.9×10^{-7} to $8.6 \times 10^{-7} \text{ m s}^{-1}$, Graham 2008). Even higher values of bedrock saturated hydraulic conductivity were observed at the Panola hillslope and in the Fudoji catchment (Uchida et al. 2003a; Tromp-van Meerveld et al. 2007). At the Panola hillslope, at least 20% of the precipitation during large rainstorms infiltrates into the granite bedrock (Tromp-van Meerveld et al. 2007). Kosugi et al. (2006) reported an annual bedrock infiltration ranging from 35 to 55% of the annual precipitation in an unchanneled 0.024-ha catchment near Fudoji (weathered granite). The effect of bedrock leakage on the subsurface flow response at the Panola hillslope was tested through virtual experiments by Tromp-van Meerveld and Weiler (2008). They concluded that bedrock leakage needed to be included to adequately simulate the SSF response to multiple storms. When bedrock leakage was omitted, recessions were too slow and the hillslope stayed too wet in between events, leading to sustained baseflow from the hillslope and an overprediction of simulated subsurface flow responses (Tromp-van Meerveld and Weiler 2008). At the same time, exfiltration from the bedrock into the soil may be a possible process at some hillslopes (Kosugi et al. 2006, 2008b). The soil–bedrock exchange processes are yet not well characterized, even though tracer experiments (Noguchi et al. 1999; Frazier et al. 2002) and bedrock groundwater monitoring documented flow within bedrock fractures (Kosugi et al. 2008a).

23.3.6 *Interrelations and Mutual Dependencies*

As pointed out in the foregoing summary of controlling factors, there are numerous interrelations among different controls of rainfall-runoff processes at the hillslope scale (e.g., Tromp-van Meerveld and McDonnell 2006c; Hopp and McDonnell 2009).

For instance, the spatial variation of soil moisture is partly due to rainfall partitioning, which is driven by both vegetation and storm characteristics. Furthermore, plant roots alter the soil structure, which, in turn, affects infiltration (Angers and Caron 1998). At the same time, hillslope hydrological processes and its controls are mutually dependent. On the one hand, climate and soil moisture control vegetation dynamics; on the other hand, vegetation controls the hillslope water balance and interacts with the atmosphere (Porporato and Rodriguez-Iturbe 2002). In other words, there is a double feedback between hydrological processes and its controls. Or, as Rodriguez-Iturbe (2000) put it, the spatial distribution of soil moisture (and probably also the saturated zone) and its evolution in time is both cause and consequence of vegetation. This also applies to interactions among hydrology, topography and soil properties. Along a toposequence, soil properties will change due to soil moisture patterns driven by topography (e.g., different weathering rates and biological activity in depressions vs. ridges). Yet, differences in soil properties cause different hydrological processes. This co-evolution of hydrology and landscape (Wagener et al. 2007) adds even more complexity to the system.

As a result of the many interrelations and mutual dependencies, identifying the first-order controls driving the rainfall-runoff response at a specific hillslope is highly challenging, especially since the impact of each control may be temporally variable. Nonetheless, we are confronted with the need to assess the hydrologic behavior of ungauged hillslopes and to develop modeling concepts. Given this nonlinear complex system, what are the hillslope characteristics we have to consider, and how do we pool the many pieces of information to assess its hydrologic response? Generally, the static “hillslope configuration” (geologic, morphologic, and pedologic properties such as soil depth and drainable porosity) determines the predisposition for certain runoff processes. This predisposition is refined through dynamic controls (input characteristics and vegetation dynamics) (e.g., Williams et al. 2008; Hopp and McDonnell 2009). Low-gradient hillslopes with deep soils, for instance, are less prone to fast SSF. There, vegetation effects such as intensity smoothing by canopies and localized stemflow will most likely not strongly modify the overall hydrologic response (Keim et al. 2006). In contrast, at high-gradient hillslopes with shallow soils, such vegetation-induced effects may be highly relevant. However, for less obvious hillslope settings, we still lack the understanding how different static and dynamic hillslope attributes interact in driving the runoff response. We propose that the concept of connectivity provides a viable framework for assessing the underlying controls of the nonlinear rainfall-runoff response at the hillslope scale.

23.3.7 Connectivity as Concept for Assessing the Rainfall-Runoff Response

In recent years, connectivity has emerged to a popular concept (Ali and Roy 2009; Michaelides and Chappell 2009). However, the term “connectivity” was often used ambiguously (e.g., Western et al. 2001; Pringle 2003; Knudby and Carrera 2005;

Bracken and Croke 2007; Lehmann et al. 2007; Michaelides and Chappell 2009), and “there is no consensus among hydrologists about what it is exactly” (Ali and Roy 2009). Similar to the definition by Stieglitz et al. (2003), we understand hydrologic connectivity at the hillslope scale as a system state, in which disconnected “hydrologically-active areas” (Uchida et al. 2001) (areas of surface flow, saturated soil patches, water-channeling subsurface features) interconnect with each other and to the stream, thus significantly contributing to storm runoff.

Many studies have proven that a key factor for surface and subsurface runoff is the connectivity of runoff-producing patches (Stieglitz et al. 2003; McNamara et al. 2005; Lehmann et al. 2007; Gomi et al. 2008). For example, Lehmann et al. (2007) applied percolation theory to the Panola hillslope, and concluded that the rainfall threshold triggering rapid hillslope out-flow seems to be related to the flow path connectivity at the next smaller spatial scale. Recently, Michaelides and Chappell (2009) recommended using connectivity to extract a simplified, universally applicable, quantifiable characteristic that transcends individual hillslope complexity.

Following this incentive, we suggest that the hillslope hydrologic response is determined by the connectivity of hydrologically-active areas on and within the slope. There are three types of hydrologically active areas: (a) areas of surface flow, (b) saturated soil patches, and (c) high-permeability features, which are live and decayed roots, animal burrows, subsurface erosion channels, bedrock fractures, and the soil skeleton. The mechanisms generating areas of surface flow and high-permeability features are straightforward (e.g., rainfall intensity exceeding the infiltration capacity, surface sealing, hydrophobicity, topsoil saturation; presence of structural features and self-reinforcement through subsurface erosion). The mechanisms producing spatially disconnected, saturated soil patches are diverse and can be summarized as follows: (1) variations in surface and bedrock topography, causing saturated conditions preferably in depressions, (2) spatially variable precipitation input due to the double-funneling effect of vegetation aboveground and belowground (interception, throughfall, stemflow, transpiration, litter layer partitioning, preferential flow along roots), and (3) variations in soil depth and drainable porosity (see Fig. 23.2). Once hydrologically active areas are established, they seem to recur during the next event, as observed for finger flow in water-repellent soils (Ritsema and Dekker 2000; Ritsema et al. 1998a, b).

We hypothesize that a significant increase in hillslope outflow is bound to the interconnection of hydrologically active areas; yet, it does not matter what kind of hydrologically active areas interconnect, or by which means they were formed. The fill-and-spill hypothesis (Tromp-van Meerveld and McDonnell 2006b) offers one explanation for the often observed nonlinearity of hillslope runoff response (bedrock depressions have to fill up before a lateral connection to the stream occurs). We believe the interconnection of high-permeability features or saturated soil patches evokes a similar response. The hillslope hydrologic behavior thus relies on the formation of an internal flow network (Tsuboyama et al. 1994; Noguchi et al. 1999; Sidle et al. 2000; Uchida et al. 2001). Such a network presumably consists of different types of hydrologically active areas linked together, e.g., flow in high-permeability features interconnected via saturated soil patches, as observed in a number of

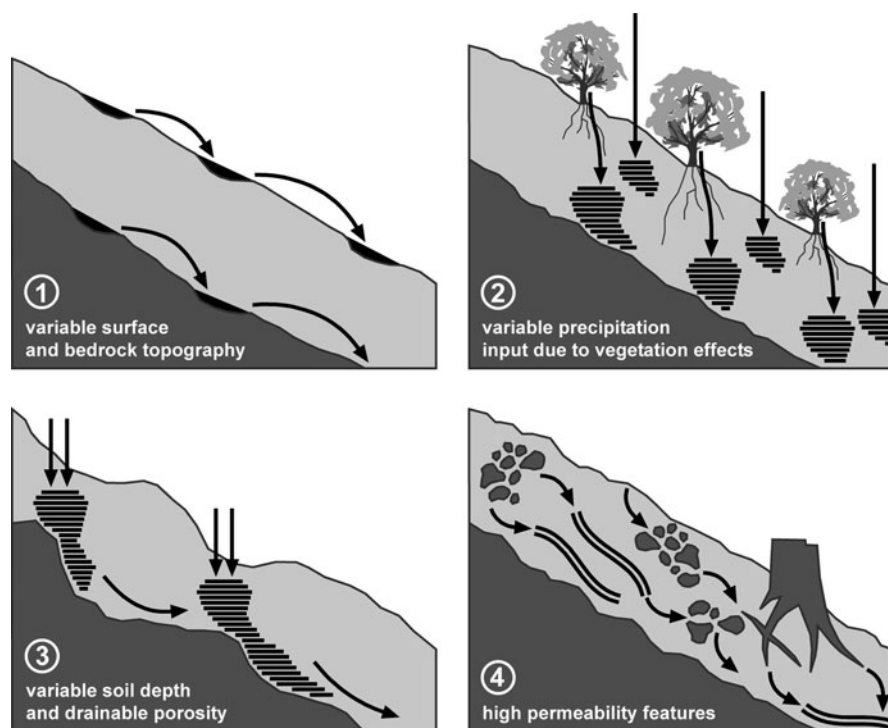


Fig. 23.2 Mechanisms generating and connecting hydrologically-active areas

experimental studies (Noguchi et al. 1999; Sidle et al. 2000; Anderson et al. 2008). In analogy to the fill-and-spill mechanism, we propose a connect-and-react mechanism: only if hydrologically active areas interconnect, a significant reaction (e.g., hillslope outflow or transport of nutrients) is triggered (see Fig. 23.3).

Clearly, this connectivity concept per se does not improve the predictions for ungauged hillslopes (unless we identify certain hillslope attributes as proxies describing the state of the internal network). However, this concept provides a holistic yet simple view on the complex mechanisms operating in and on hillslopes, thereby subsuming the diversity of mechanisms driving hillslope runoff. It could thus serve as a perceptual framework to develop new model structures helping to predict the hydrological behavior of hillslopes.

23.4 New Directions and Research Avenues

The immense heterogeneity and complexity of hydrological processes observed in numerous environments and at different spatial scales represents a great challenge for advances in hydrology (McDonnell et al. 2007). In this context, the call for a

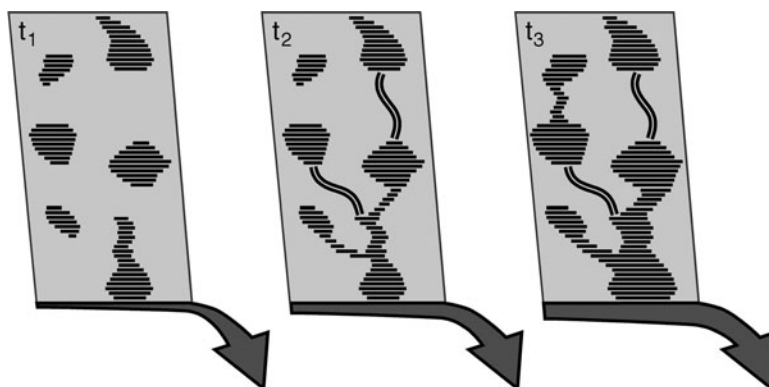


Fig. 23.3 Connect-and-react mechanism: We hypothesize that only if hydrologically active areas interconnect, a significant reaction (e.g., hillslope outflow or transport of nutrients) is triggered; yet, it does not matter what kind of hydrologically active areas interconnect, by which means they were formed, or by which mechanisms they connect

unifying hydrological theory has recently evolved (Sivapalan 2005; McDonnell et al. 2007; Troch et al. 2009). It is stressed to achieve a holistic understanding of the complex hydrologic system and its numerous interaction and feedback processes, thereby explicitly accepting landscape heterogeneity as dominant control. We thus need to systematically examine the first-order controls of hydrological processes rather than focusing on the idiosyncrasies of specific environments (Sivapalan 2005; Uchida et al. 2006; McDonnell et al. 2007; Wagener et al. 2007; Hopp et al. 2009; Troch et al. 2009). Troch et al. (2009) argue that, despite the tremendous heterogeneity of landscape properties, there is structure and organization across spatial and temporal scales. Understanding how and why this heterogeneity occurs may help to improve hydrologic predictability (Troch et al. 2009). In this respect, it was proposed to search for the reasons of heterogeneity and to identify the underlying principles or natural laws governing the hydrologic response (Sivapalan 2005; McDonnell et al. 2007). Regarding hillslope hydrology, what are feasible steps moving towards this great vision?

23.4.1 *New Dimensions with Regard to Content*

More generally, a holistic understanding of the complex hydrologic system strongly relies on an interdisciplinary perspective. As pointed out before, there are numerous interrelations among different controlling factors driving the hillslope hydrologic response, especially with regard to the soil–plant–atmosphere continuum. Hence, research aimed at identifying underlying principles or natural laws of hydrological processes needs the synergies from many disciplines, such as ecology, biology, biogeochemistry, pedology, geomorphology, and meteorology (Sivapalan 2005;

Lin et al. 2006; Troch et al. 2009). The necessity of interdisciplinary work and its mutual stimulation is expressed by the recent development of conjoined disciplines such as ecohydrology (Rodriguez-Iturbe 2000), hypedopedology (Lin 2003), or hydro-geomorphology (Sidle and Onda 2004). We believe that a general process understanding of hillslope hydrology is especially reliant on how vegetation patterns and dynamics influence the hillslope water balance. In this respect, ecohydrology, which “seeks to describe the hydrologic mechanisms that underlie ecologic patterns and processes” (Rodriguez-Iturbe 2000) will play a significant role (e.g., Bond 2003; Hannah et al. 2004; Barnard 2009). As pointed out in the section on vegetation as controlling factor, plants affect hillslope soil moisture patterns and runoff formation in numerous ways. It is interesting though how many experimental and modeling studies neglect these effects. Numerous studies analyzing the hillslope runoff response do not pay special attention to the vegetation growing on the examination area, even if pipe flow is observed (e.g., McDonnell 1990; Woods and Rowe 1996; Kim et al. 2004, 2005; Fujimoto et al. 2008). Then, there are hillslope studies systematically investigating dominant controlling factors by intercomparing slopes differing in certain attributes (soil, bedrock, topography) under varying boundary conditions (rainfall amount and intensity, antecedent wetness); however, only under one type of land use (e.g., Kienzler and Naef 2008a, b), or poorly differentiating between grassland and forest, without specifying what type of forest, grassland, etc. (Scherrer et al. 2007). This shows that vegetation is not thought to be a first-order control. In addition, artificial sprinkling experiments suppress the rainfall partitioning effect of vegetation since water is applied close to the surface and not above the canopy (e.g., Markart et al. 2004; Scherrer et al. 2007; Nordmann et al. 2009). Virtual experiments by Keim et al. (2006), however, showed that the role of interception on SSF may be highly relevant. The influence of different root systems on flow routing in hillslopes has also been rarely considered (e.g., Nordmann et al. 2009), despite growing evidence that preferential flow along roots may override pore-scale controls on infiltration (e.g., Angers and Caron 1998; Bundt et al. 2001; Buttle and McDonald 2002; Bachmair et al. 2009; Lange et al. 2009). In summary, numerous experimental and modeling studies investigating the hillslope runoff response disregard plant effects. At the same time, there are many studies examining how certain tree and stand characteristics influence throughfall or stemflow (see Chap. 21). How this, in turn, affects hillslope runoff is yet seldom inferred, leaving room for interpretations on the actual implications. This gap should be addressed by future ecohydrological hillslope studies.

An idea originating from ecology that has been fruitfully transferred to hydrology is the focus on spatial patterns (Rodriguez-Iturbe 2000). The search for spatial patterns seems promising since it may provide a link to the underlying mechanisms (Rodriguez-Iturbe 2000; Sivapalan 2005). Studies correlating the hydrologic response with quantifiable landscape characteristics (e.g., Broxton et al. 2009) represent valuable attempts in this direction. In this respect, the potential power of optimality principles applied to hydrology shall be pointed out (Reggiani et al. 1998; Kleidon and Schymanski 2008; Schymanski et al. 2008; Rasmussen et al. 2011). Following optimality approaches, which are based on the

principle of maximum entropy production, complex systems far from thermodynamic equilibrium organize in a way that the rate of entropy production is maximized in steady state (Kleidon and Schymanski 2008). Spatial heterogeneity is thus not the result of some random process but the consequence of certain mass and energy fluxes (Schulz et al. 2006; Rasmussen et al. 2011). Further ideas on explaining hydrological patterns in relation to soil and landscape features may come from hydro-pedology (Lin 2003; Lin et al. 2006). According to Troch et al. (2009), identifying ways to quantify the implications of soil development and soil structure on hydrological processes seems promising. In combination with approaches predicting pedogenesis from more readily available information on climate and landscape position, this may be a valuable step toward improving the hydrologic predictability in ungauged regions (Troch et al. 2009).

23.4.2 Research Design

The highlighted new dimensions of hillslope hydrology have several implications for the research design of future hillslope studies. We need a unifying framework for experimental set-ups, analysis strategies and modeling concepts to clear the way for identifying underlying principles of hydrological processes at the hillslope scale.

In general, an improved process conceptualization needed for the development and testing of hydrological models relies on intercomparing processes at different hillslopes to abstract from the ubiquitous spatial and temporal variability. However, there are several obstacles in this respect. First, due to the numerous ways of experimental hillslope set-ups (e.g., trenched vs. nontrenched slopes, different hillslope sizes, artificial sprinkling vs. natural events), examined response metrics (e.g., total hillslope outflow vs. pipe flow, spatiotemporal resolution of internal response characteristics), and measurement techniques (e.g., different types and conditions of tracer application) comparability among different hillslope experiments is not given. Second, many experimental studies operate on rather small hillslope segments; it is questionable whether this size is suitable for deriving general answers (e.g., size of trench or hillslope outflow face: 4 m (Scherrer et al. 2007), 5 m (Kim et al. 2004, 2005), 8 m (Retter 2007; Kienzler and Naef 2008a), 9 m (Anderson et al. 2009), 10 m (McGuire et al. 2007), 16 m (Newman et al. 1998), 20 m (Tromp-van Meerveld and McDonnell 2006a), and 60 m (Woods and Rowe 1996; Graham 2008). Additionally, we need a large number of gauged hillslopes in several contrasting environments to draw meaningful conclusions. Yet, experimental hillslope studies are expensive and labor intensive. We hence need to figure out which response characteristics we have to measure for sufficiently describing the system. This goes hand in hand with the development and application of novel measurement techniques, such as ground penetrating radar or electrical resistance tomography for identifying subsurface structures (Holden et al. 2002; Cassiani et al. 2006; Schulz et al. 2006; Uhlenbrook et al. 2008; Markart et al. 2009),

soil water isotope extractions (Wassenaar et al. 2008), water isotope interactions with vegetation (Barnard 2009), wireless sensor techniques (Trubilowicz et al. 2009), and incremental stream discharge measurements (Szeftel et al. *in review*). Overall, we have to develop a sort of “minimum requirements” concept for hillslope studies to make them comparable and suitable for model validation and rejection. Furthermore, we should generally aim for intercomparison studies as research design instead of characterizing just one hillslope in detail (Uchida et al. 2005; McDonnell et al. 2007). Such intercomparison studies were successfully applied by a number of authors at the hillslope scale (e.g., Asano et al. 2002, 2004; Scherrer and Naef 2003; Uchida et al. 2004, 2005, 2006; Scherrer et al. 2007; Kienzler and Naef 2008a, b; Nordmann et al. 2009) and catchment scale (e.g., Molenat et al. 2008; Broxton et al. 2009; Hrachowitz et al. 2009; Tetzlaff et al. 2009). However, it is complicated to find natural hillslopes where certain controlling factors can be isolated, and attempts to artificially construct large-scale hillslopes as reported by Hopp et al. (2009) are an exception. Thus, virtual experiments (Weiler and McDonnell 2004) represent an effective framework for advancing our process understanding, as recent studies prove (e.g., Keim et al. 2006; Weiler and McDonnell 2006; Dunn et al. 2007; Tromp-van Meerveld and Weiler 2008; Hopp and McDonnell 2009). Additionally, we should think about ways facilitating data sharing as basis for intercomparison studies, as postulated by Soulsby et al. (2008), which would also benefit model evaluation and improvement.

In conclusion, there have been many attempts to go beyond simply “cataloging” the complexity of rainfall-runoff processes and to develop a more holistic picture of the factors driving the hillslope hydrologic response. Furthermore, the suggested directions regarding research gaps and design of future experimental and modeling studies may be a further step toward the vision of a unified hydrologic theory.

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